

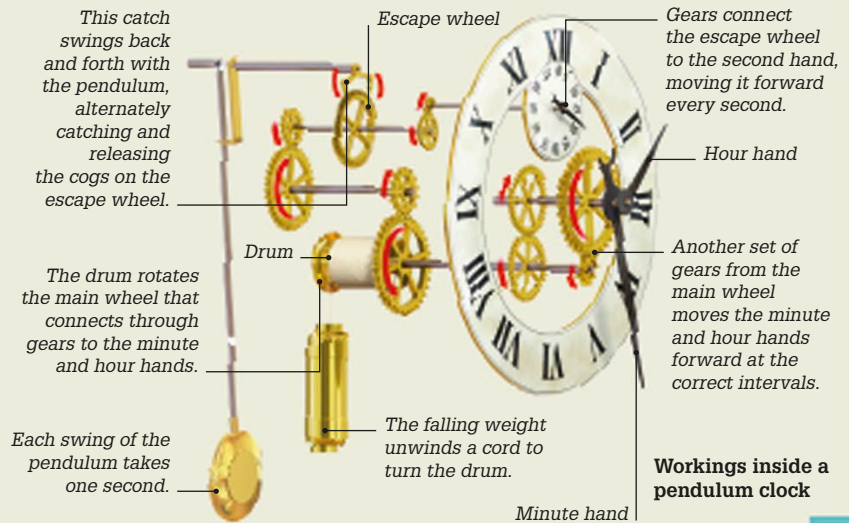


Inner dial shows the signs of the Zodiac.

Outer dial shows the 24 hours of the day in Roman numerals.

How pendulum clocks work

A pendulum clock can keep excellent time because the rate at which the pendulum swings can be precisely controlled. Inside the clock, the pendulum regulates the rate at which a drum turns, via a catch on the escape wheel. A weight unwinds the cord from around the drum and turns the gears, which are connected to the minute and hour hands. The weight takes eight days to unwind the cord and then the cord must be rewound with a key.



World's first caesium atomic clock, 1955

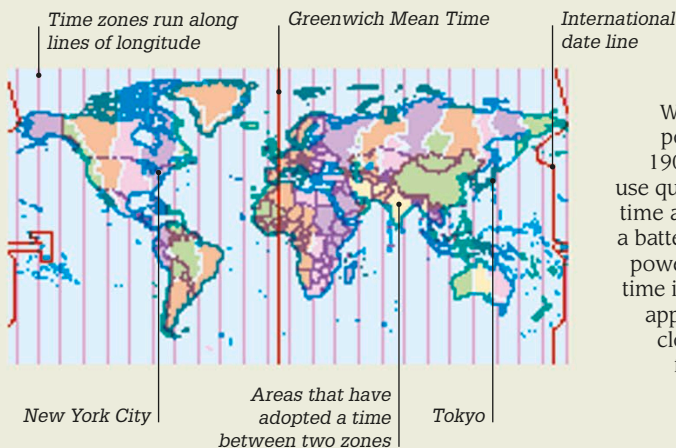
Atomic clocks

Atomic clocks are the most accurate way of keeping time yet known. They keep time by counting the high-frequency waves emitted by energized atoms in elements such as caesium. A caesium clock will not gain a second in 1 million years.

Time zones

Since 1884, to standardize the time around the world, the globe has been divided into 24 time zones.

Times are measured from Greenwich Mean Time, or GMT, with each zone either hours ahead or behind GMT. For example, New York City is five hours behind GMT and Tokyo is nine hours ahead.



Modern watches

Wristwatches have been popular since the early 1900s. Today's watches use quartz crystals to keep time and are powered by a battery, or even by solar power. They display the time in analog (the way it appears on a traditional clock face) or in digital form, according to the wearer's preference.



Early digital watch, 1970s

1884

Greenwich Mean Time became the global time standard after an international conference adopted the Greenwich meridian in the UK as the Prime Meridian (line of 0° longitude).



Marker for the Prime Meridian Line at Greenwich, UK

1927

The first quartz clock, driven by the natural electricity generated by a rapidly vibrating quartz crystal, was built at the Bell Telephone Laboratories, New York City.

1949

The first atomic clock was built in Washington D.C. In 1967, a second was redefined as the time that elapses during 9,192,631,770 cycles of radiation from a caesium-133 atom.

1970s

Watches and clocks began to show the time in digital form using light-emitting diode (LED) or liquid-crystal display (LCD) instead of a traditional (analog) clock face.